

Annex 178B

(normative)

ILT and RTS functions and support for autonomous path startup (APSU) for electrical and optical interfaces

178B.1 Scope

This annex defines functions to support autonomous path startup (APSU) for Physical Layer implementations that include one or more inter-sublayer links (ISLs) (see 178B.3) with data rate of 200 Gb/s or higher per lane.

178B.2 Overview

APSU is the coordinated, orderly initialization of all ISLs and sublayers in a path (see Figure 178B–1). APSU is facilitated by the ready-to-send (RTS) function and, where required, the inter-sublayer link training (ILT) function. APSU is described in 178B.4.

The RTS function is an internal function of an AUI component or a PMD, collectively referred to as interfaces (see 178B.3), that propagates the readiness of each ISL to transition to DATA mode. The RTS function propagates the RTS status between peer and adjacent interfaces. RTS status indicates whether an ISL is ready to transition to DATA mode, or not. The RTS status is transferred between adjacent interfaces via the service interface primitives, and between peer interfaces via the ILT function or another PMD-specific function. The RTS function is described in 178B.6 and 178B.10.

For PAM4 ISLs, the ILT function is an internal function of an AUI component or a PMD that controls the establishment of communication with a peer interface within an inter-sublayer link (ISL). The ILT function provides optimization of data signaling and coordinates the transition to DATA mode on an ISL. The ILT function is defined in 178B.7.

For coherent ISLs, the ILT function is not defined.

NOTE—AUI components and PMDs that are not specified to support the ILT function or the RTS function as defined in this annex may include equivalent capabilities, however this is beyond the scope of this standard.

Terminology used in this annex is provided in 178B.3.

178B.3 Definitions

For the purpose of this annex, the following definitions apply. Refer to 1.4 for terms not defined in this subclause.

Adjacent interface

An interface whose service interface is connected with this interface via one or more intermediate sublayers. See Figure 178B–2.

- AUI component
- An AUI component is either a C2C component (e.g., see 176C.3) or a C2M component (e.g., see 176D.3). In a device with two AUI components the upper AUI component is the one facing toward the MAC sublayer and the lower AUI component is the one facing toward the Medium.
- Interface
- Unless qualified otherwise, interface is either an AUI component or a PMD. See Figure 178B–3.
- Inter-sublayer link (ISL)
- An ISL is either a pair of AUI components and the AUI channel between, or a pair of PMDs (in different PHYs) and the medium between. See Figure 178B–1.
- Path
- The sequence of ISLs and intermediate sublayers providing the connectivity between two Reconciliation Sublayers (RSs). See Figure 178B–1.
- Peer interface
- The interface at the other end of the ISL. In the case where the ISL is between a pair of PMDs, this term is equivalent to link partner. See Figure 178B–2.

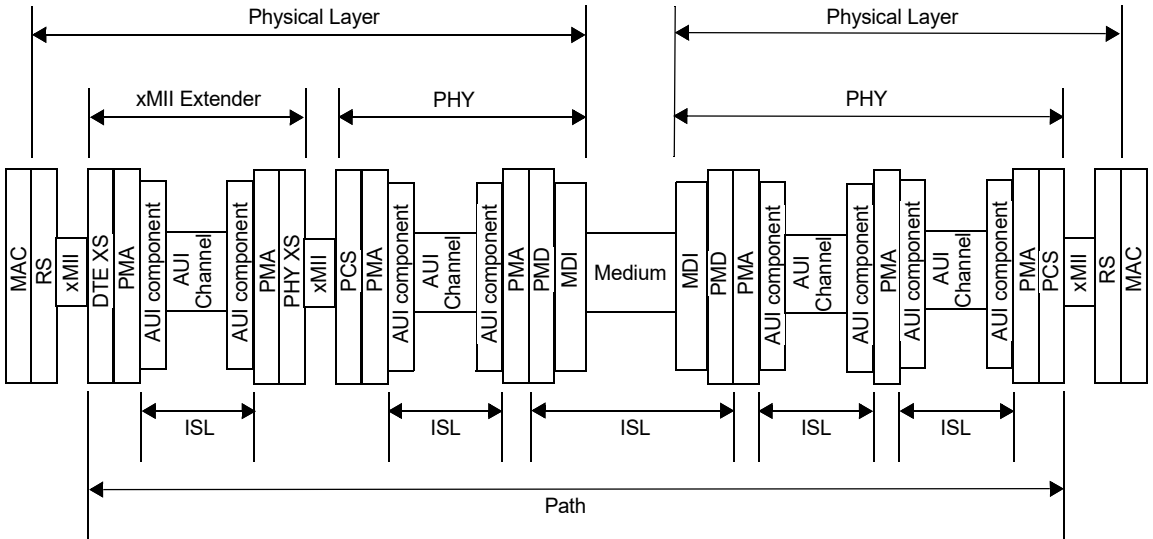


Figure 178B–1—ISL and Path

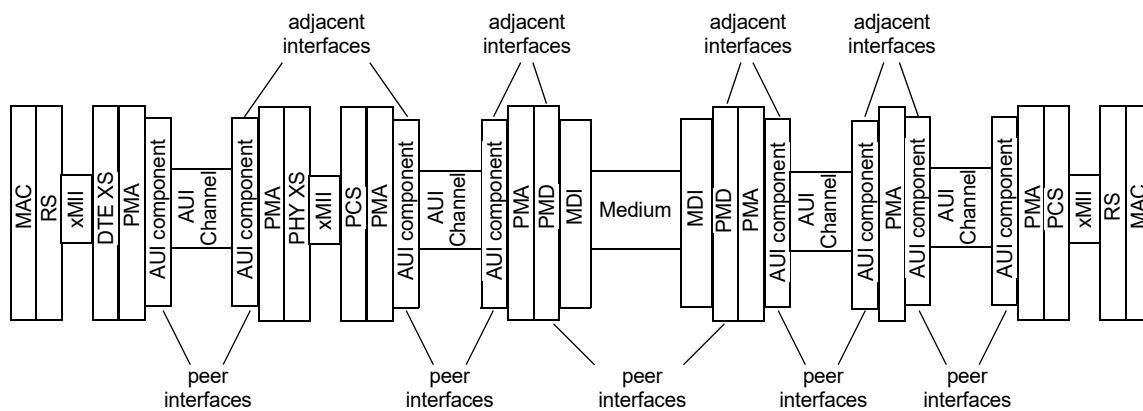


Figure 178B-2—Peer and adjacent interfaces

178B.4 Autonomous path startup (APSU)

APSU is not intrinsically a function, it is an externally observable behavior resulting from establishing communication through all sublayers in the path in both directions. Support for APSU is defined as follows:

- An ISL can be activated using APSU if the two interfaces and the associated sublayers (e.g. PMA, Inner FEC), implement the RTS function (see 178B.6 and 178B.10) and, where required, the ILT function (see 178B.7), or have equivalent functions.
- A PHY can be activated using APSU if every ISL and sublayer within the PHY can be activated using APSU.
- An xMII Extender can be activated using APSU if every ISL and sublayer within it can be activated using APSU.
- A Physical Layer can be activated using APSU if the PHY and xMII Extender (if implemented) can be activated using APSU.
- A path can be activated using APSU if the Physical Layer at each end can be activated using APSU.

A path that does not meet the above criteria is not defined to support APSU. An ISL, PHY, Physical Layer, or path that cannot be activated using APSU may be activated using management or other means beyond the scope of this annex.

APSU is the result of SIGNAL_OK for every sublayer interface reaching the OK status. This occurs when:

- Each sublayer is receiving and transmitting data of the expected format.
- Communication is established in both directions of each ISL. For AU component and PMDs using the RTS and ILT functions defined in 178B.6 and 178B.7, the variables local_rts and remote_rts are both true. For PMDs using the RTS function defined in 178B.10, the variables c_local_rts and c_remote_rts are both true.
- End-to-end communication between the RS sublayers is established.

178B.5 RTS and ILT functions within a PAM4 interface

The RTS function and ILT function within an interface are illustrated in Figure 178B-3.

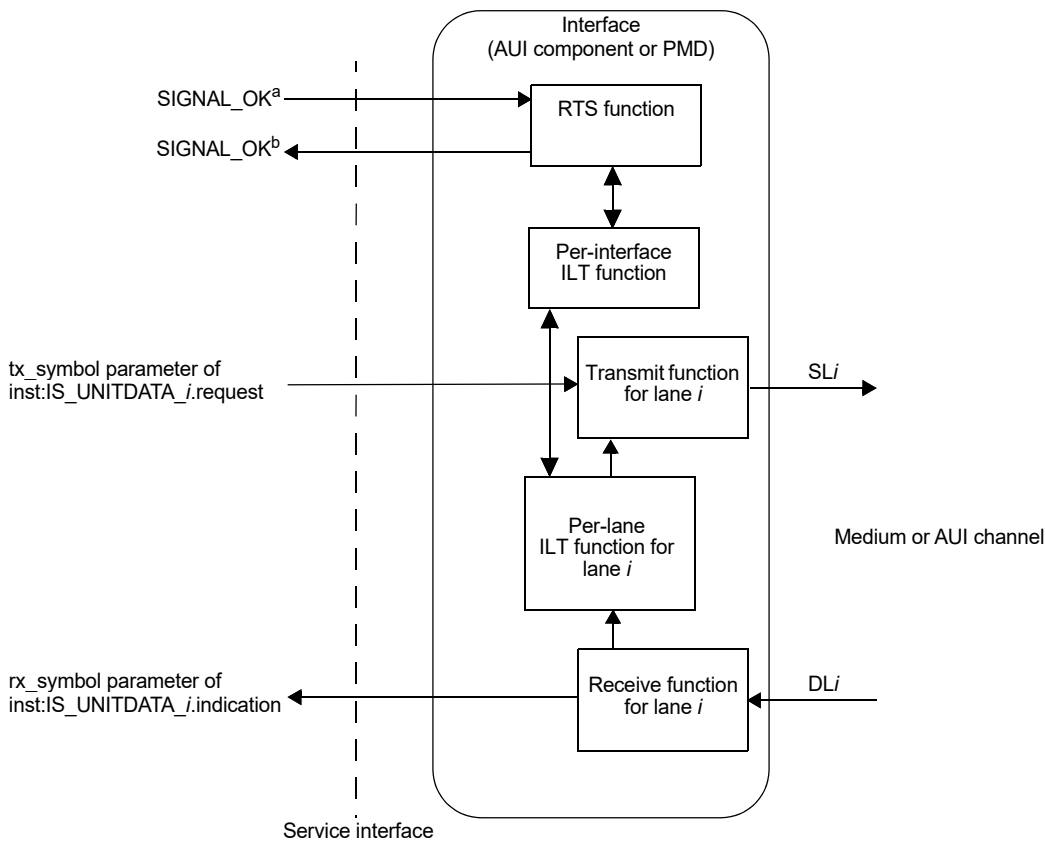
The ILT function within an interface is composed of:

- one per-interface ILT function
- n per-lane ILT functions, one for each of n lanes associated with the interface

The RTS function uses inputs from the following sources to permit the ILT function to transition to DATA mode:

- the adjacent interface and signal indication logic (SIL) in one or more sublayers between the local interface and its adjacent interface (via the service interface)
- the peer interface (via the ILT function)
- the per-interface ILT function

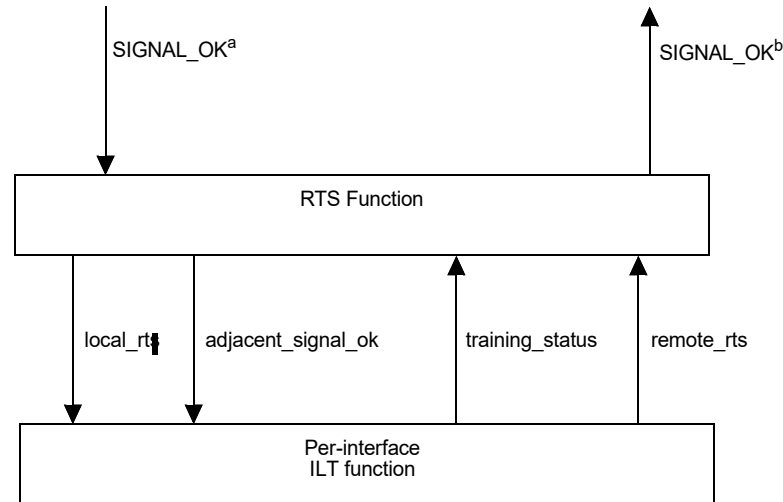
The signals between the RTS function and ILT function are illustrated in Figure 178B–4.



^a From an adjacent interface, a DTE XS or a PCS next to an RS
^b To an adjacent interface, a DTE XS or a PCS next to an RS

SL_i = source lane i
 DL_i = destination lane i

Figure 178B–3—RTS and ILT functions within interfaces



^a From an adjacent interface, a DTE XS or a PCS next to an RS
^b To an adjacent interface, a DTE XS or a PCS next to an RS

Figure 178B-4—Signaling between RTS and ILT functions

178B.6 RTS function for PAM4 interfaces

The RTS function controls the switchover of the transmitter clock source to its DATA mode clock (when necessary) and facilitates the indication of the local interface readiness to switch to DATA mode (*local_rts*) to other interfaces (peer and adjacent). The RTS update state diagram (Figure 178B-9) and its associated variables define the process used for the clock source transition. The operation of the RTS function is different based upon the state of *mr_training_enable* and whether the interface is implemented in a retimer (see 178B.9).

An overview of the behavior of the RTS function is as follows:

- Initially, the *local_rts* variable is set false.
- Wait for the adjacent interface to be ready for DATA mode, as indicated by the *SIGNAL_OK* parameter from the adjacent interface being equal to READY or OK.
- If the ILT function is operating in TRAINING mode (*mr_training_enable* = true), wait for local ISL to complete its adaptation, as indicated by *training_status* equal to READY or OK.
- If the interface is implemented in a retimer that uses the recovered clock in DATA mode, initiate the swap to the recovered clock from the adjacent interface.
- Set the *local_rts* variable to true, indicating readiness to move to DATA mode.

The state of the *local_rts* variable is provided to the peer interface via the ILT function (Figure 178B-4). The state of the *rts_status* variable is provided by the interface to the adjacent interface via the service interface *SIGNAL_OK* parameter.

There is no specified timeout when waiting for either the *local_rts* or the *remote_rts* variables to change.

178B.7 ILT function for PAM4 interfaces

The ILT function facilitates the establishment of communication between the peer interfaces of an ISL, including clock and data recovery and the transfer of the RTS status of the transmitter. The per-interface ILT function (see 178B.7.1) provides an aggregate status of all the per-lane ILT functions (see 178B.7.1) to the RTS function and indicates the readiness of the transmitter to send data (*local_rts*) to the per-lane ILT functions.

The ILT function has three modes of operation based on the *tx_mode* variable (see 178B.8.3.1):

- TRAINING mode (*tx_mode* = TRAINING): training frames are transmitted to the peer interface
- LOCAL_PATTERN mode (*tx_mode* = LOCAL_PATTERN): a pattern is transmitted to the peer interface
- DATA mode (*tx_mode* = DATA): data is passed from the channel to the service interface and vice versa

When the ILT function is not in DATA mode, the transmitter ignores the input service interface *IS_UNITDATA* primitive and the ILT transmit function locally generates the data that is sent to its peer interface.

TRAINING mode is used when training is enabled (*mr_training_enable* = true). This mode utilizes a training protocol, which enables the two peer interfaces to communicate with each other through transmission of fixed-length training frames (see 178B.7.3). The training protocol provides methods for each interface of an ISL to request changes of the peer interface transmitter state, provides status in response to requests received, provides status of the local receiver state, and coordinates the transition to DATA mode.

LOCAL_PATTERN mode is used when training is disabled (*mr_training_enable* = false). This mode provides a method to indicate the readiness of the transmitter to transition to DATA mode to the peer interface using a locally generated pattern, as specified in the clause or annex that defines the interface.

DATA mode is entered when the ILT function has successfully completed its operations. If training is enabled (*mr_training_enable* = true), when transitioning from TRAINING mode to DATA mode the state of certain transmitter parameters that were derived by the training protocol are retained (e.g. precoder on/off).

The training control state diagram (Figure 178B–10a and Figure 178B–10b) and its associated variables are used by both TRAINING mode and LOCAL_PATTERN mode of operation. The state diagrams in Figure 178B–11 and Figure 178B–12 and their associated variables, apply only when training is enabled (*mr_training_enable* = true).

Recovery from the ISL_FAIL state in the training control state diagram (Figure 178B–10b) requires management to assert *mr_restart_training* or reset; the timing of this action is implementation dependent and beyond the scope of this standard.

178B.7.1 Per-interface ILT function

The per-interface ILT function monitors all of the independent per-lane ILT functions, provides summary status information to the RTS function, and relays RTS states to the per-lane ILT functions.

Variables that are managed by the ILT function include: *remote_rts*, *isl_ready*, *training_status*. See 178B.8.2.1.

178B.7.2 Per-lane ILT function

The per-lane ILT function is responsible for the locally generated data that is transmitted while not in DATA mode and for the adaptation processes of the received signal. Each lane of the interface runs an independent instance of the per-lane ILT function.

When in TRAINING mode, the frame format specified in 178B.7.3 is used by the receiver to configure its peer interface transmitter (see 178B.7.4). There are two formats of the training protocol frames defined, E1 and O1 (see 178B.7.4 and 178B.7.5). The clause or annex that defines the interface specifies which format is used. The transmitter parameters available for modification by the training protocol depends upon the interface type.

When in LOCAL_PATTERN mode, the pattern transmitted is specified in the clause or annex that defines the interface. An interface indicates RTS to its peer interface by exiting the transmit disable state (tx_disable = false). The transmitter and receiver operate independently of each other. The transmitter parameters are configured through management control variables. A receiver adapts to the incoming signal regardless of the local transmitter state.

The per-lane ILT function is defined by 178B.7.3 through 178B.7.10 and the state diagrams in Figure 178B–10a, Figure 178B–10b, Figure 178B–11, and Figure 178B–12.

178B.7.3 Training frame structure

The training frame is a sequence of PAM2 or PAM4 symbols. The structure of a training frame is shown in Figure 178B–5.

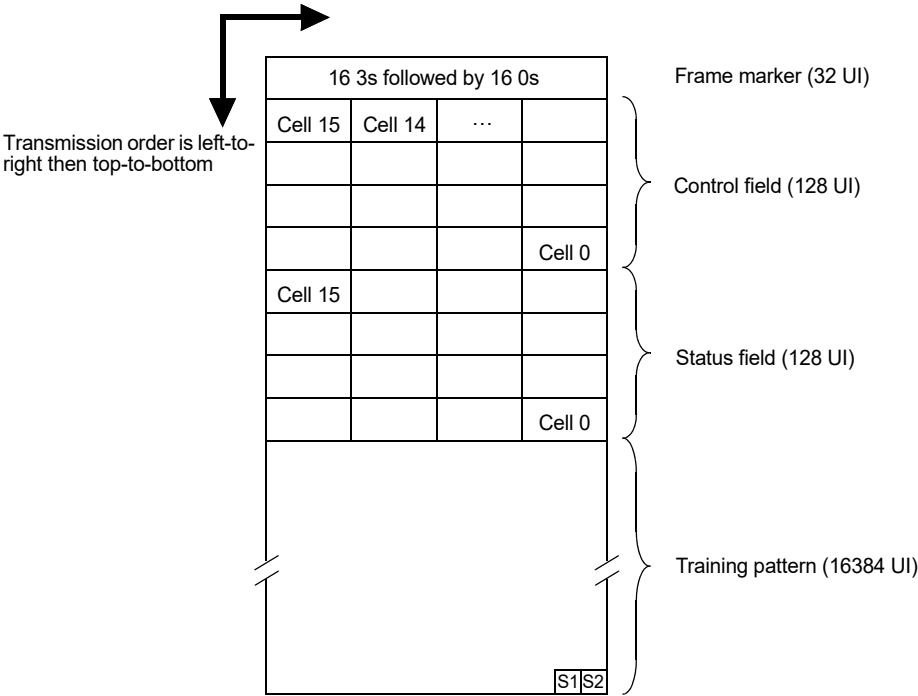


Figure 178B–5—Training frame structure

178B.7.3.1 Frame marker

Training frames are delimited by a specific sequence of PAM4 symbols. The training frame marker shall be a run of 16 consecutive “3” symbols followed by a run of 16 consecutive “0” symbols. This sequence is not found in the control field, status field, or training pattern and it uniquely identifies the beginning of a training frame.

178B.7.3.2 Control and status fields

The control field is composed of 16 bits with the structure defined in 178B.7.4. The status field is composed of 16 bits with the structure defined in 178B.7.5.

Each bit of the control and status fields is sent as a differential Manchester encoded (DME) cell, where each cell is eight unit intervals in length. The specific rules for this encoding follow.

- A transition from 0 to 3 or from 3 to 0 occurs at the start of each cell.
- A transition from 0 to 3 or from 3 to 0 at the midpoint of a cell, i.e., four unit intervals from the transition at the beginning of the cell, corresponds to a logical one.
- The absence of a transition at the midpoint of a cell corresponds to a logical zero.

The control field is transmitted immediately after the frame marker. The status field is transmitted immediately after the control field. Within each field, the order of transmission is from bit 15 to bit 0.

When a training frame is received, if a violation of the DME encoding rules is detected within the control field or the status field, the contents of both fields in that frame are ignored.

Two formats are defined for the control and status fields, E1 and O1. Which training frame format is used is specified by the clause or annex that defines the interface.

178B.7.3.3 Training pattern

The training pattern is the result of a training pattern generator equivalent to the structure shown in Figure 178B–6.

There are three training pattern generator functions: synchronous PRBS13, free-running PRBS13, and free-running PRBS31.

The contents of the last two symbols of the training frame payload (S1 and S2 in Figure 178B–5) depends on the training pattern. For synchronous PRBS13 these two symbols are “0” and for free-running PRBS13 and free-running PRBS31 these two symbols are taken from the training pattern generator output.

The training pattern is either PAM2 or PAM4 modulated. For PAM4 modulation, precoding is either enabled or disabled (see 178B.7.3.4).

The training pattern selector is set to synchronous PRBS13 and the modulation to PAM2 upon entry to the SEND_TRAINING state of the training control state diagram (Figure 178B–10a).

Exiting TRAINING mode requires both sides to use PAM4 modulation, thus the modulation and precoding request and status are both required to have values other than PAM2.

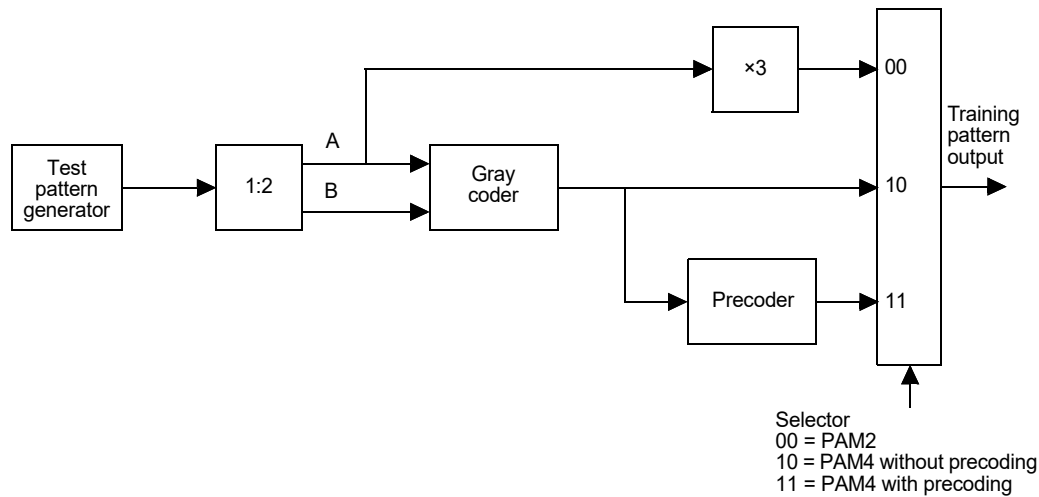


Figure 178B-6—Training pattern generator

178B.7.3.3.1 Synchronous PRBS13 function

The synchronous PRBS13 function for each lane shall implement four generator polynomials. The polynomial used in each lane i is selected by the variable identifier_i .

At the start of the training pattern in each training frame in each lane i , the state of the PRBS13 generator shall be set to the value of the variable prbs13_seed_i . A value of all zeros is not valid.

Table 178B-1 specifies the default identifier, the corresponding polynomial, and the recommended default value of prbs13_seed_i for each lane i , as well as the first 13 symbols of the training pattern for PAM2, PAM4 without precoding, and PAM4 with precoding, created using the default polynomial and seed.

A sample implementation of the PRBS13 generator for $\text{identifier}_i = 0$ is shown in Figure 178B-7.

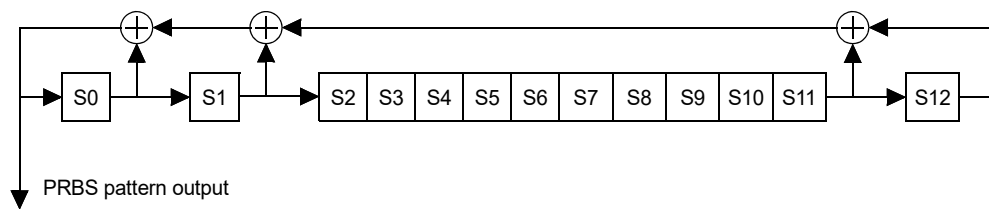


Figure 178B-7—Test pattern PRBS13 generator ($\text{identifier}_i = 0$)

Table 178B–1—Training pattern default polynomials and seeds

<i>i</i>	Default identifier	Polynomial, $G(x)$	Default prbs13_seed_ <i>i</i> ^a	Initial output, PAM2	Initial output, PAM4 without precoding	Initial output, PAM4 with precoding
0	0 ^b	$1 + x + x^2 + x^{12} + x^{13}$	0000010101011	0030330330000	1031320220111	1301200200101
1	1	$1 + x^2 + x^3 + x^7 + x^{13}$	0111010000011 ^c	3330003000330	2220002101230	2022220132031
2	2	$1 + x^2 + x^4 + x^8 + x^{13}$	1001000101100	0303333033030	1212332133031	1102120121301
3	3	$1 + x^2 + x^5 + x^9 + x^{13}$	0100010000010	3330300030330	2231210121221	2032013201110
4	0 ^b	$1 + x + x^2 + x^{12} + x^{13}$	1111110100110	3030000303303	3030001313212	3122223012011
5	1	$1 + x^2 + x^3 + x^7 + x^{13}$	1000111011100 ^c	3033030003030	3032020002021	3120022220023
6	2	$1 + x^2 + x^4 + x^8 + x^{13}$	0000001101000	3300303000300	2300212111300	2131102323000
7	3	$1 + x^2 + x^5 + x^9 + x^{13}$	0011000100111	3333000333030	2232000322031	2033131202210

^a The leftmost bit in the sequence corresponds to the initial value of S0 and the rightmost bit corresponds to the initial value of S12 in Figure 178B–7.

^b This training pattern is equivalent to the PRBS13Q test pattern defined in 120.5.11.2.1.

^c The default seed values for *i* = 1 and *i* = 5 are different from the default seed values in Table 162–10a.

178B.7.3.3.2 Free-running PRBS13 function

A set of four free-running PRBS13 generator polynomials are defined to minimize correlated interference between physical lanes during training. The free-running PRBS13 generator for each lane shall implement each of four generator polynomials defined in Table 178B–1. The polynomial used in each lane *i* is selectable by identifier_*i*. The default identifier for each lane is the same as that shown in Table 178B–1. A sample implementation of the free-running PRBS13 generator for identifier_*i* = 0 is shown in Figure 178B–7.

The initial value of the free-running PRBS13 generator shall not be all zeros. It may be any other value. For eight-lane interfaces the same polynomial is used for two lanes. The two generators shall be configured such that their relative offsets are large enough to make adjacent lanes uncorrelated within the length of the training frame. For example, this may be achieved by initialization with different seeds or with the same seed at different times.

The free-running PRBS13 generator operates independently of the training frame and is not stopped or reset while training is in progress and this mode is selected. As a result, the state of the free-running PRBS13 generator at the beginning of the training pattern is different in subsequent training frames. The marker, control, and status DME portions of the training frame (see 178B.7.3) periodically override the free-running PRBS13 generator output (for 288 UI every 16672 UI).

178B.7.3.3.3 Free-running PRBS31 function

The free-running PRBS31 generator for each lane shall implement the PRBS31 pattern defined by Equation (49-2) and shown in Figure 49–9.

The initial state of the free-running PRBS31 generator shall not be all zeros. It may be any other value. The free-running PRBS31 generators in a multi-lane interface shall be configured such that their relative offsets

are large enough to make adjacent lanes uncorrelated within the length of the training frame. For example, this may be achieved by initialization with different prbs31_seed_i ($i = 0$ to 7) or with the same seed at different times.

The free-running PRBS31 generator operates independently of the training frame and is not stopped or reset while training is in progress and this mode is selected. As a result, the state of the free-running PRBS31 generator at the beginning of the training pattern is different in subsequent training frames. The marker, control, and status DME portions of the training frame (see 178B.7.3) periodically override the free-running PRBS31 generator output (for 288 UI every 16672 UI).

178B.7.3.4 Training pattern modulation and coding

The training pattern generator generates three different training patterns. The construction of each training pattern begins by demultiplexing the test pattern generator output into pairs of bits {A, B} where A corresponds to the 1st, 3rd, 5th, etc. bits output by the test pattern generator and B corresponds to the 2nd, 4th, 6th, etc. bits output by the test pattern generator. There is a new pair of bits each unit interval which implies that the test pattern generator is required to generate bits at twice the signaling rate. Given these pairs of bits, the three different training patterns correspond to three values of the training pattern selector in Figure 178B–6: PAM2, PAM4 without precoding, and PAM4 with precoding.

When the training pattern selector is set to PAM2, the training pattern is the sequence of PAM2 symbols derived by mapping the A bits from the test pattern generator such that logical 0 is transmitted as 0 and logical 1 is transmitted as 3.

When the training pattern selector is set to PAM4 without precoding, the training pattern is the sequence of PAM4 symbols derived by Gray coding the {A, B} pairs as specified for outputs in 120.5.7.1.

When the training pattern selector is set to PAM4 with precoding, the training pattern is the sequence of PAM4 symbols derived by Gray coding the {A, B} pairs as specified for outputs in 120.5.7.1 and precoding the result as specified for outputs in 135.5.7.2.

For synchronous PRBS13, at the beginning of each training pattern the precoder state is set to 0 such that $P(j-1) = 0$ in Equation (135–1) for the first PAM4 symbol of the training pattern. For free-running PRBS13 and PRBS31, it is recommended that the PRBS test pattern generator and the precoder are initialized simultaneously with the seed (for the PRBS test pattern generator) and $P(j-1) = 0$ (for the precoder).

178B.7.4 Control field structure

The structure of the control field for the E1 format shall be as shown in Table 178B–2 and for the O1 format as shown in Table 178B–3.

Table 178B-2—Control field structure for the E1 format

Bit(s)	Name	Description
15:14	Reserved	Transmit as 0, ignore on receipt
13:11	Initial condition request	13 12 11 1 1 1 = Reserved 1 0 1 = Preset 6 0 1 1 = Preset 5 0 0 1 = Preset 4 1 1 0 = Preset 3 1 0 0 = Preset 2 0 1 0 = Preset 1 0 0 0 = Individual coefficient control
10	Not ready to send	1 = Not ready to send 0 = Switch to data when training is completed
9:8	Modulation and precoding request	9 8 1 1 = PAM4 with precoding 1 0 = PAM4 without precoding 0 1 = Reserved 0 0 = PAM2
7	Reserved	Transmit as 0, ignore on receipt
6:5	Training pattern request	6 5 1 1 = free-running PRBS31 1 0 = Reserved 0 1 = free-running PRBS13 0 0 = synchronous PRBS13
4:2	Coefficient select	4 3 2 1 0 0 = Reserved 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$ 0 1 x = Reserved
1:0	Coefficient request	1 0 1 1 = No equalization 1 0 = Decrement 0 1 = Increment 0 0 = Hold

Table 178B–3—Control field structure for the O1 format

Bit(s)	Name	Description
15:11	Reserved	Transmit as 0, ignore on receipt
10	Not ready to send	1 = Not ready to send 0 = Switch to data when training is completed
9:8	Modulation and precoding request	9 8 1 1 = PAM4 with precoding 1 0 = PAM4 without precoding 0 1 = Reserved 0 0 = PAM2
7	Reserved	Transmit as 0, ignore on receipt
6:5	Training pattern request	6 5 1 1 = free-running PRBS31 1 0 = Reserved 0 1 = free-running PRBS13 0 0 = synchronous PRBS13
4:0	Reserved	Transmit as 0, ignore on receipt

178B.7.4.1 Initial condition request

Initial condition request only applies for the E1 format. The initial condition request bits are used to select one of the up to six predefined transmitter equalizer configurations (presets) specified in the clause or annex that defines the interface. When the initial condition request bits are 000, the coefficient select and coefficient request bits are used to update the transmitter equalizer configuration.

178B.7.4.2 Not ready to send

The not ready to send bit carries the inverse value of the local_rts variable (see 178B.8.2.1).

178B.7.4.3 Modulation and precoding request

The modulation and precoding bits are used to request that the peer interface transmitter modulates and codes the transmitted training pattern using one of the options defined in 178B.7.3.4.

178B.7.4.4 Training pattern request

The training pattern bits are used to request that the peer interface transmitter transmits one of the training patterns defined in 178B.7.3.3.1, 178B.7.3.3.2, and 178B.7.3.3.3.

178B.7.4.5 Coefficient select

Coefficient select only applies for the E1 format. The coefficient select bits are used to identify the coefficient that is the target of a coefficient request. The value of the field is the two's complement encoding of the coefficient index. For example, binary value 111 corresponds to a coefficient index of –1.

178B.7.4.6 Coefficient request

Coefficient request only applies for the E1 format. The coefficient request bits are used to change the value of the coefficient specified by the coefficient select bits. A coefficient is changed by incrementing or decrementing its value or by setting it to NO EQUALIZATION. The NO EQUALIZATION value is 1 for $c(0)$ and 0 for $c(-3)$, $c(-2)$, $c(-1)$, and $c(1)$.

178B.7.5 Status field structure

The structure of the status field for the E1 format shall be as shown in Table 178B–4 and for the O1 format as shown in Table 178B–5.

Table 178B–4—Status field structure for the E1 format

Bit(s)	Name	Description
15	Receiver ready	1 = Training is complete and the receiver is ready for data 0 = Request for training to continue
14	Reserved	Transmit as 0, ignore on receipt
13:12	Training pattern status	13 12 1 1 = free-running PRBS31 1 0 = Reserved 0 1 = free-running PRBS13 0 0 = synchronous PRBS13
11:10	Modulation and precoding status	11 10 1 1 = PAM4 with precoding 1 0 = PAM4 without precoding 0 1 = Reserved 0 0 = PAM2
9	Receiver frame lock	1 = Frame boundaries identified 0 = Frame boundaries not identified
8	Initial condition status	1 = Updated 0 = Not updated
7	Parity	Even parity bit
6	Reserved	Transmit as 0, ignore on receipt
5:3	Coefficient select echo	5 4 3 1 0 0 = Reserved 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$ 0 1 x = Reserved
2:0	Coefficient status	2 1 0 1 1 1 = Reserved 1 1 0 = Coefficient at limit and equalization limit 1 0 1 = Reserved 1 0 0 = Equalization limit 0 1 1 = Coefficient not supported 0 1 0 = Coefficient at limit 0 0 1 = Updated 0 0 0 = Not updated

Table 178B–5—Status field structure for the O1 format

Bit(s)	Name	Description
15	Receiver ready	1 = Training is complete and the receiver is ready for data 0 = Request for training to continue
14	Reserved	Transmit as 0, ignore on receipt
13:12	Training pattern status	13 12 1 1 = free-running PRBS31 1 0 = Reserved 0 1 = free-running PRBS13 0 0 = synchronous PRBS13
11:10	Modulation and precoding status	11 10 1 1 = PAM4 with precoding 1 0 = PAM4 without precoding 0 1 = Reserved 0 0 = PAM2
9	Receiver frame lock	1 = Frame boundaries identified 0 = Frame boundaries not identified
8	Reserved	Transmit as 0, ignore on receipt
7	Parity	Even parity bit
6:0	Reserved	Transmit as 0, ignore on receipt

178B.7.5.1 Receiver ready

The receiver ready bit is used to signal the local receiver state to the peer interface. When this bit is 1, it indicates that the local receiver has completed training and is prepared to receive data (`local_rx_ready = true`). When this bit is 0, it indicates that the local receiver is requesting that training continue (`local_rx_ready = false`).

178B.7.5.2 Training pattern status

The training status bits are encoded to convey the value of `local_tp_mode` (see 178B.8.3.1).

178B.7.5.3 Modulation and precoding status

The modulation and precoding bits are encoded to convey the value of `local_mc_mode` (see 178B.8.3.1).

178B.7.5.4 Receiver frame lock

The receiver frame lock bit is set to 1 when the receiver has identified the training frame marker position and is in a state where the response time requirements specified in 178B.7.10 are met.

Receiver frame lock is set to 0 when the variable training is false, and is not set to 1 until training and `local_tf_lock` are both true and the response time requirements specified in 178B.7.10 can be met.

178B.7.5.5 Initial condition status

Initial condition status only applies for the E1 format. The initial condition status bit acknowledges the initial condition request bits received from the peer interface. The acknowledgment reflects the value of `ic_sts` resulting from the procedure described in 178B.7.8.1.

178B.7.5.6 Parity

The parity bit is calculated based on the other bits in the control field and status field to create even parity for these fields. Even parity ensures that the transmitted control and status fields (see 178B.7.3.2) are DC balanced. This field is ignored on receipt.

178B.7.5.7 Coefficient select echo

Coefficient select echo only applies for the E1 format. The coefficient select echo bits acknowledge the coefficient select bits received from the peer interface. When a change to the coefficient select bits is detected, the coefficient select echo bits are updated to represent the same coefficient index. This serves as a confirmation to the peer interface that subsequent coefficient requests will act on the targeted coefficient. The coefficient select echo bits reflect the value of the `k` variable generated by the coefficient update state diagram (Figure 178B–12).

178B.7.5.8 Coefficient status

Coefficient status only applies for the E1 format. The coefficient status bits acknowledge the coefficient request bits received from the peer interface. The coefficient status bits reflect the value of the `coef_sts` variable generated by the coefficient update state diagram (Figure 178B–12).

178B.7.6 Training frame lock

Training frame lock shall be achieved by detecting either the frame marker or the inverted frame marker in consecutive training frames as specified in Figure 178B–11.

178B.7.7 Polarity detection and correction

When training starts for each lane, the variable `polarity_correction` is set to false. If inverted frame markers are detected during the frame lock process, the `polarity_correction` variable is set to true.

If `polarity_correction` is true and `local_tf_lock` is true, the lane input shall be corrected by mapping the received PAM4 symbols 0, 1, 2, and 3 to PAM4 symbols 3, 2, 1, and 0, respectively.

NOTE—When training is disabled, support of polarity detection and correction is implementation dependent.

The state of the `polarity_correction` variable persists after training completes, correcting the polarity of the data received when `tx_mode` = DATA.

178B.7.8 Equalization control

Equalization control is only available for the E1 format.

When the training control state diagram (Figure 178B–10a) is in the TRAIN_LOCAL state, the interface may request its peer interface to change the transmitter equalization coefficients, either to predefined initial conditions or by individual coefficient control. The criteria for initiating such requests are implementation dependent.

When the training control state diagram (Figure 178B–10a) is in either the TRAIN_LOCAL or TRAIN_REMOTE states, the interface shall respond to requests received from the peer interface.

When the training control state diagram (Figure 178B–10a and Figure 178B–10b) is in any state other than TRAIN_LOCAL or TRAIN_REMOTE, the interface shall not send any request to the peer interface and shall ignore requests from the peer interface.

A new individual coefficient update request or initial condition update request is not initiated until after the prior request has completed.

178B.7.8.1 Initial condition setting request process

A request to change the initial condition of the peer interface transmitter shall be made by using the following procedure:

- a) Set the initial condition request bits (178B.7.4.1) to the desired predefined transmitter equalizer configuration (preset) and the coefficient request bits (178B.7.4.6) to HOLD.
- b) Wait until the initial condition status bits (178B.7.5.5) indicate UPDATED and the coefficient status bits (178B.7.5.8) indicate NOT UPDATED or COEFFICIENT NOT SUPPORTED.
- c) Set the initial condition request bits (178B.7.4.1) to INDIVIDUAL CONTROL and the coefficient request bits (178B.7.4.6) to HOLD.
- d) Wait until both the initial condition status bits (178B.7.5.5) and the coefficient status bits (178B.7.5.8) indicate NOT UPDATED.

178B.7.8.2 Initial condition setting response process

The handling of incoming requests is specified by the coefficient update state diagram (Figure 178B–12). The behavior of the UPDATE_IC function shall be consistent with the following algorithm.

```

if ic_req = INDIVIDUAL CONTROL
    ic_sts = NOT UPDATED
else if CHECK_REQ(ic_req)
    Set coefficients according to ic_req
    ic_sts = UPDATED
    coef_sts = NOT UPDATED
else
    ic_sts = UPDATED
    coef_sts = COEFFICIENT NOT SUPPORTED
end if

```

The variables ic_req, ic_sts and coef_sts are defined in 178B.8.3.1. The transmitter equalizer coefficients presets shall be within the ranges specified in the clause or annex that defines the interface for the selected preset. The function CHECK_REQ is defined in 178B.8.3.2.

178B.7.8.3 Coefficient update request process

A request to change an individual equalizer coefficient of the peer interface transmitter shall be made by using the following procedure:

- a) In the transmitted control field, set the initial condition request bits (178B.7.4.1) to INDIVIDUAL CONTROL, and set the coefficient select bits (178B.7.4.5) and coefficient request bits (178B.7.4.6) to the desired values. This may be done in one step (all fields updated in the same training frame) or sequentially.
- b) Wait until the received coefficient status bits (178B.7.5.8) do not indicate NOT UPDATED and the coefficient select echo bits (178B.7.5.7) indicate the requested coefficient select value.

- c) If the subsequent request is to change the same coefficient or request a new initial condition, set the coefficient request bits (178B.7.4.6) to HOLD and wait until the received coefficient status bits (178B.7.5.8) indicate NOT UPDATED.

178B.7.8.4 Coefficient update response process

The handling of incoming requests is specified by the coefficient update state diagram (Figure 178B–12). The behavior of the UPDATE_C(k) function shall be consistent with the following algorithm.

```

if k in k_list
    if coef_req = INCREMENT
        ck_ask = c(k) + ck_stp
    else if coef_req = DECREMENT
        ck_ask = c(k) – ck_stp
    else if coef_req = NO EQUALIZATION
        if k = 0
            ck_ask = 1
        else
            ck_ask = 0
        end if
    else
        ck_ask = c(k)
    end if
    if ck_ask > ck_max
        c(k) = ck_max
        if CHECK_EQ(ck_ask,k)
            coef_sts = COEFFICIENT AT LIMIT AND EQUALIZATION LIMIT
        else
            coef_sts = COEFFICIENT AT LIMIT
        end if
    else if ck_ask < ck_min
        c(k) = ck_min
        if CHECK_EQ(ck_ask,k)
            coef_sts = COEFFICIENT AT LIMIT AND EQUALIZATION LIMIT
        else
            coef_sts = COEFFICIENT AT LIMIT
        end if
    else if CHECK_EQ(ck_ask,k)
        coef_sts = EQUALIZATION LIMIT
    else
        c(k) = ck_ask
        coef_sts = UPDATED
    end if
else
    coef_sts = COEFFICIENT NOT SUPPORTED
end if

```

The variables coef_req, coef_sts, and k are defined in 178B.8.3.1. The following additional variables and functions are used.

- c(k)
Variable that contains the current value of the coefficient $c(k)$.
- ck_ask
Variable that contains the value of $c(k)$ that would result from the requested update.

ck_min	Variable that contains the minimum supported value of $c(k)$.
ck_max	Variable that contains the maximum supported value of $c(k)$.
ck_stp	Variable that represents the magnitude of the change in $c(k)$ for one step up or one step down from its current value. The value is implementation dependent but within the range specified by the clause or annex that defines the interface.
k_list	The set of valid transmitter equalizer coefficient indices as specified by the clause or annex that defines the interface.
CHECK_EQ(ck_ask,k)	Compares the transmitter steady-state voltage that would result from setting transmit equalization coefficient $c(k)$ value to ck_ask, while keeping all other coefficients unchanged, against the transmitter equalization capability and coefficient ranges (see 179.9.4.2.5 as an example). Returns true if the resulting combination of coefficients would exceed the transmitter equalization capability or coefficient ranges. Otherwise returns false.

178B.7.9 Training pattern setting

When a change to the received modulation and precoding request bits or the training pattern request bits is detected, the transmitted training pattern (see 178B.7.3.3) is set accordingly. To confirm that the change to the format of the training pattern was completed, the local_mc_mode variable is set to the value of the received modulation and precoding request bits and the local_tp_mode variable to the value of the received training pattern request bits. local_mc_mode and local_tp_mode are encoded in status fields (see 178B.7.5.2 and 178B.7.5.3).

178B.7.10 Handshake timing

Changes to the configuration of the transmitter equalizer or the transmitted training pattern occur at any point between the time the request for the new configuration is received and the time that the request is acknowledged.

When the receiver frame lock bit in the status field of transmitted training frames is set to 1, the time from the receipt of a new request to the acknowledgment of that request shall be less than 20 ms. It is recommended that the average response time be less than 2 ms. A new request is defined to be a received training frame whose control field differs from the control field of the preceding training frame. An acknowledgment is defined as the first transmitted training frame that contains a status field encoding that is an appropriate response for the requested action. For example, a change in the coefficient select bits in the control field would be acknowledged by a change in the coefficient select echo bits of the status field. Similarly, a change in the pattern request bits in the control field would be acknowledged by a change in the pattern status bits in the status field. All timing measurements are referred to a common reference point within the training frame (e.g., the 3 to 0 transition in the training frame marker).

The timing of requests and acknowledgment is illustrated in Figure 178B–8.

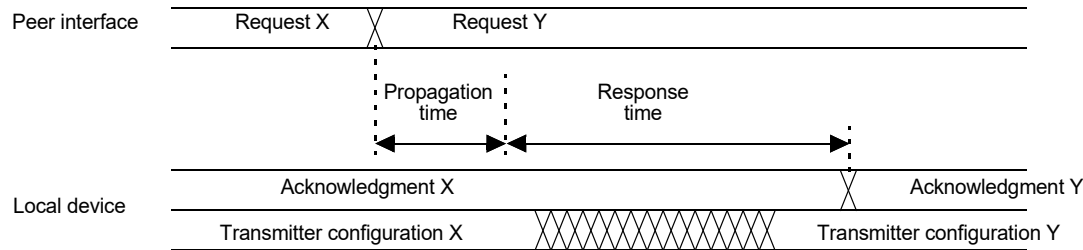


Figure 178B-8—Transmitter update timing

178B.8 State diagrams for PAM4 interfaces

This subclause is composed of detailed state diagram processes, including the associated definition of variables, functions, counters, and timers.

178B.8.1 State diagram conventions

The notation used in the state diagrams follows the conventions of 21.5.

All timers operate as described in 14.2.3.2 with one addition. A timer is reset and stops counting upon entering a state where “stop x_timer” is stated.

Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

178B.8.2 RTS and ILT per-interface state diagram definitions

An interface implements one instance of the RTS update state diagram, and the set of associated variables, functions, counters and timers defined in this subclause.

178B.8.2.1 Variables

adjacent_signal_ok

Enumerated variable derived from the value of the SIGNAL_OK parameter of the input IS_SIGNAL primitive on the service interface. It takes one of the following values: IN_PROGRESS, READY, OK, FAIL.

NOTE—For an interface below a PMA or Inner FEC (AUI component or PMD), the adjacent_signal_ok variable is derived from the SIGNAL_OK parameter of the IS_SIGNAL.request primitive on the service interface above the interface. For an interface above a PMA (AUI component), the adjacent_signal_ok variable is derived from the SIGNAL_OK parameter of the IS_SIGNAL.indication primitive on the service interface below the interface.

adjacent_intf_rx_ready

Boolean variable that is true when adjacent_signal_ok is either READY or OK and false otherwise. If there is no adjacent interface this variable is always true.

isl_ready

Boolean variable that is set to true when local_rx_ready and remote_rx_ready are true for all lanes of the interface, and to false otherwise.

local_rts

Boolean variable that indicates that an interface is ready to send data. Its value is set by the RTS update state diagram (Figure 178B–9). The logical-NOT of this variable is encoded as the not ready to send bit in the control field of transmitted training frames.

mr_adj_los_mode

Enumerated variable assigned by management that can take one of the following values: FAIL, SQUELCH, or LOCAL_PATTERN. The default value is FAIL.

mr_restart

Boolean variable used by management to restart the RTS and ILT functions.

NOTE — There is no specified time limit for the RTS and ILT functions.

mr_training_enable

Boolean variable that is set by management. When it is true, training is enabled on the interface. When it is false, training is not enabled on the interface.

NOTE — Training is enabled by default for interfaces that support it. ILT requires that mr_training_enable is set to the same value at both ends of each ISL. Disabling training on ISLs that support it might compromise performance and APSU.

remote_rts

Boolean variable that indicates the value of local_rts variable in the peer interface. If mr_training_enable is true then remote_rts is set to true when the received training frames on all lanes of the interface have the not ready to send bit (see 178B.7.4.2) set to zero, otherwise it is set to false. If mr_training_enable is false then remote_rts is always true.

reset

Boolean variable that controls the global resetting of the RTS and ILT state machines. It is true whenever a reset is necessary including when initiated by PMA_reset for AUI components, PMD_reset for PMDs, and during power on, and it is false otherwise.

rts_status

Enumerated variable that indicates the status of the RTS function. This variable is mapped to the SIGNAL_OK parameter of the output IS_SIGNAL primitive on the service interface. It is assigned one of the following values as follows:

FAIL: training_status is FAIL

OK: training_status is OK

READY: remote_rts is true and training_status is READY

IN_PROGRESS: training_status is IN_PROGRESS or remote_rts is false and training_status is READY

NOTE — For an interface below a PMA or Inner FEC (AUI component or PMD), rts_status maps to the SIGNAL_OK parameter of the IS_SIGNAL.indication primitive on the service interface above the interface. For an interface above a PMA (AUI component), rts_status maps to the SIGNAL_OK parameter of the IS_SIGNAL.request primitive on the service interface below the interface.

rx_fail_on_los

Boolean variable that indicates the receiver requires a restart of the ILT function in order to recover from a loss of signal event (rx_ok becomes false) when the training control state diagram (Figure 178B–10b) is in the PATH_UP state. The value of this variable is implementation dependent. Default for this variable is true.

Editor's note: The rx_fail_on_los variable was added per Draft 3.1 comment R1-248. The register bit for rx_fail_on_los is defined as a read-only ability bit. Therefore defining a default is not meaningful. Proposals to address this are encouraged.

training_status

Enumerated variable that indicates the status of the ILT function. This variable is assigned one of

the following values as follows:

FAIL: Any lane_training_status is FAIL

OK: All lane_training_status are OK

READY: All lanes have lane_training_status of TRAINED or OK with at least one lane being TRAINED

IN_PROGRESS: At least one lane has lane_training_status of IN_PROGRESS and no lanes have lane_training_status of FAIL

tx_lp_on_adj_los_ability

Boolean variable that indicates the interface supports sending LOCAL_PATTERN when the training control state diagram (Figure 178B–10b) is in the PATH_UP state and local_rts becomes false. The value of this variable is implementation dependent.

tx_sq_on_adj_los_ability

Boolean variable that indicates the interface supports SQUELCH when the training control state diagram (Figure 178B–10b) is in the PATH_UP state and local_rts becomes false. The value of this variable is implementation dependent.

uses_recovered_clock

Boolean variable set to true to indicate that the interface uses a clock recovered by an adjacent interface (see 178B.9) to drive its output in DATA mode (tx_mode = DATA, see 178B.8.3.1). Otherwise it is set to false.

178B.8.2.2 Functions

USE_TX_CLOCK(source)

Selects the clock source for the transmitter. The source parameter takes one of the values:

local: use a local clock to drive the transmitter output.

recovered: use a clock recovered by the adjacent interface receiver (see 178B.9) to drive the transmitter output.

NOTE—The details of the clock recovery and forwarding architecture are beyond the scope of this standard.

178B.8.2.3 Timers

forward_rts_timer

This timer is started when the RTS update state diagram (Figure 178B–9) enters the SWITCH_CLOCK state. The terminal count of forward_rts_timer is between 10 ms and 90 ms.

178B.8.2.4 State diagram figure

The RTS update state diagram (Figure 178B–9) sets the value of the local_rts variable and it shall be implemented for each interface.

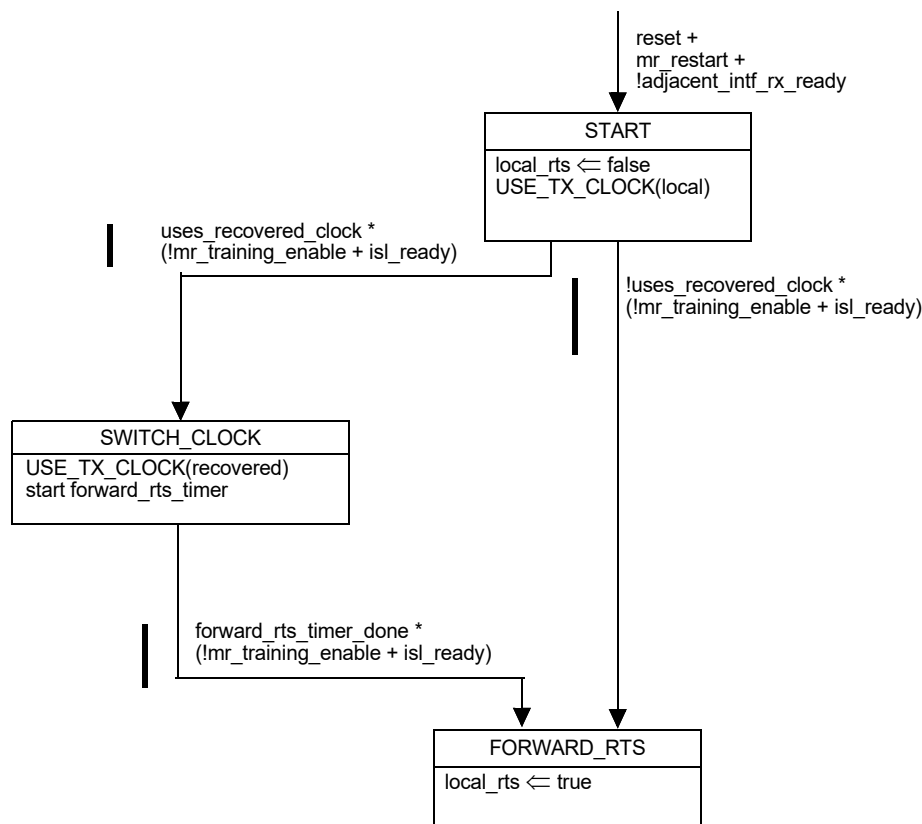


Figure 178B-9—RTS update state diagram

178B.8.3 ILT per-lane state diagram definitions

Each lane of an interface using the E1 format implements one instance of each of the training control, the training frame lock and the coefficient update state diagrams.

Each lane of an interface using the O1 format implements one instance of each of the training control and the training frame lock state diagrams.

If the PATH_READY state in the training control state diagram (Figure 178B-10b) is entered with local_mc_mode set to “PAM4 with precoding”, then the interface shall cause the PMA or Inner FEC to transmit all subsequent data on the corresponding lane with precoding (see 176.7.1.2) and otherwise cause the PMA or Inner FEC to transmit all subsequent data on the corresponding lane without precoding.

If the PATH_READY state in the training control state diagram (Figure 178B-10b) is entered with the modulation and coding status of the status field of the received training frames set to “PAM4 with precoding”, then the interface shall inform the PMA or Inner FEC that all subsequently received data on the

corresponding lane includes precoding (see 176.7.1.2) and otherwise inform the PMA or Inner FEC that all subsequently received data on the corresponding lane does not include precoding.

178B.8.3.1 Variables

adj_los_mode

Enumerated variable that controls the path taken out of the ADJ_LOS state in the training control state diagram (Figure 178B–10b). It is set as follows:

LOCAL_PATTERN: when mr_adj_los_mode is LOCAL_PATTERN and

tx_lp_on_adj_los_ability is true.

SQUELCH: when mr_adj_los_mode is SQUELCH and tx_sq_on_adj_los_ability is true.

FAIL: Otherwise

coef_req

Enumerated variable derived from the “coefficient request” bits from the control field of the received training frames (see 178B.7.4). This variable is assigned one of the following values:

HOLD, DECREMENT, INCREMENT, NO EQUALIZATION.

coef_sel

Variable derived from the “coefficient select” bits from the control field of the received training frames (see 178B.7.4). It is assigned a signed integer value that is the two’s complement interpretation of the bits.

coef_sts

Enumerated variable that is assigned one of the following values: NOT UPDATED, UPDATED, COEFFICIENT AT LIMIT, COEFFICIENT NOT SUPPORTED, EQUALIZATION LIMIT, COEFFICIENT AT LIMIT AND EQUALIZATION LIMIT. The value is assigned by the UPDATE_C(k) function and coefficient update state diagram (Figure 178B–12), and then encoded in the status field of transmitted training frames via the ENCODE_STS function.

ic_req

Enumerated variable that holds the initial condition setting to be applied to the local transmitter. It is assigned any of the values assigned to ic_sel, or the value INITIALIZE. The value of ic_req is set by the coefficient update state diagram (Figure 178B–12).

ic_sel

Enumerated variable derived from the “initial condition request” bits from the control field of the received training frames (see 178B.7.4). This variable is assigned one of the following values: INDIVIDUAL CONTROL, PRESET 1, PRESET 2, PRESET 3, PRESET 4, PRESET 5, PRESET 6.

ic_sts

Enumerated variable that is assigned one of the following values: NOT UPDATED, UPDATED. The value of ic_sts is set by the coefficient update state diagram (Figure 178B–12).

inverse_marker_valid

Boolean variable that is set to true when the candidate frame marker matches the inverse of the frame marker pattern defined in 178B.7.3.1 and is set to false otherwise.

k

Variable that stores the most recent value of coef_sel. The value of k is set by the coefficient update state diagram (Figure 178B–12).

lane_training_status

Enumerated variable that indicates the status of the per-lane ILT function. This variable is assigned one of the following values: IN_PROGRESS, TRAINED, OK or FAIL. The value of lane_training_status is set by the training control state diagram (Figure 178B–10a and Figure 178B–10b).

local_mc_mode	1
Enumerated variable that controls the training pattern modulation and coding, and is assigned one of the following values: PAM2, PAM4 without precoding, PAM4 with precoding. The value of local_mc_mode is initially set to PAM2 by the training control state diagram (Figure 178B–10a) when entering the SEND_TRAINING state and then set based on requests from the peer interface according to section 178B.7.9.	2 3 4 5 6
local_rx_ready	7
Boolean variable that indicates the state of the receiver. Its value is set by the training control state diagram (Figure 178B–10a).	8 9 10
local_tf_lock	11
Boolean variable that is true when mr_training_enable is true and the training frame marker position has been identified on a lane of the interface. The value of local_tf_lock is set by the training frame lock state diagram (Figure 178B–11).	12 13 14 15
local_tp_mode	16
Enumerated variable that controls the choice of the transmitted training pattern (see 178B.7.3.3) and is assigned one of the following values: synchronous PRBS13, free-running PRBS13, free-running PRBS31. The value of local_tp_mode is initially set to synchronous PRBS13 by the training control state diagram (Figure 178B–10a) when entering the SEND_TRAINING state and then set based on requests from the peer interface according to section 178B.7.9.	17 18 19 20 21
marker_valid	22
Boolean variable that is set to true when the candidate frame marker matches the frame marker pattern defined in 178B.7.3.1 and is set to false otherwise.	23 24 25
max_recovery_events	26
Integer variable that controls the maximum allowed number of transitions into the RECOVERY state in the training control state diagram (Figure 178B–10a). A value of zero allows unlimited number of transitions. The value of this variable is implementation dependent.	27 28 29
max_wait_timer_duration	30
Integer variable value that sets the duration of the max_wait_timer (see 178B.8.3.3) in seconds. The default value is specified by the clause or annex that defines the interface.	31 32 33
new_marker	34
Boolean variable that is true when a new candidate frame marker is available for testing. A new frame marker is available for testing when the training frame lock process has accumulated 32 consecutive symbols starting at the candidate frame start position. The value of new_marker is set by the training frame lock state diagram (Figure 178B–11).	35 36 37 38
polarity_correction	39
Boolean variable that is set to true when an inverted marker is detected. Otherwise it is set to false.	40 41
remote_mc_mode	42
Enumerated variable that corresponds to the “modulation and precoding status” bits in the status field of received training frames. It may be assigned one of the following values: PAM2, PAM4 with precoding, PAM4 without precoding.	43 44 45
remote_rx_ready	46
Boolean variable that indicates the value of local_rx_ready on a lane of the peer interface. If mr_training_enable is true and training is true, it is derived from the receiver ready bit of the status field of received training frames on the corresponding lane of the interface. Otherwise it is set to true.	47 48 49 50 51
remote_tf_lock	52
Boolean variable that reflects the value of local_tf_lock on a lane of the peer interface. The value of remote_tf_lock is set to true if mr_training_enable is true and the receiver frame lock bit of the	53 54

status field of the received training frame is 1 (see Table 178B–4 and Table 178B–5). Otherwise, it is set to false.

rx_ok

Boolean variable that is set to true when the receiver on a lane of the interface has determined that the peer interface transmitter is transmitting the expected signal format, that the remote transmit has been optimized if training = true, that the local receive equalizers have been optimized, and that no further adjustments are required for normal data transmission. The specific conditions for setting this variable to true or false are implementation dependent. However, it shall be set to false when an input signal corresponding to tx_disable = true in the peer is received for 100 ms (the minimum duration of the quiet_timer) or longer.

slip_done

Boolean variable that is true when the next candidate frame sync position can be tested. It is set to true when the SLIP function completes and is set to false upon entering the RESET_COUNT state of the training frame lock state diagram (Figure 178B–11).

tf_offset

Boolean variable that is true when receiving one full training frame (16 672 symbols, see 178B.7.3) following the recent frame start position, such that a new frame marker is expected. The value of tf_offset is set by the training frame lock state diagram (Figure 178B–11).

training

Boolean variable that is true when ILT is in progress using training frames (see 178B.7.3). The value of training is set by the training control state diagram (Figure 178B–10a and Figure 178B–10b).

training_failure

Boolean variable that is true when training fails to complete. The value of training_failure is set by the training control state diagram (Figure 178B–10a).

tx_disable

Boolean variable that controls the transmitter output on the lane. It is set by the training control state diagram (Figure 178B–10a and Figure 178B–10b). When it is true, the transmitter output on the lane is disabled. When it is false, tx_mode controls the content of the transmitter output on the lane.

tx_mode

Enumerated variable that controls the content of the transmitter output of the lane when tx_disable is false. It is set by the training control state diagram (Figure 178B–10a and Figure 178B–10b) to one of the values:

TRAINING: transmit training frames (see 178B.7.3)

LOCAL_PATTERN: transmits a pattern specified by the clause or annex that defines the interface

FILL: transmit either training frames or LOCAL_PATTERN

DATA: transmit data from the PMA or Inner FEC

178B.8.3.2 Functions

CHECK_REQ(ic_req)

This function compares the value of ic_req against the list of specified presets for the interface, and returns true if the value is supported and false otherwise.

ENCODE_STS

Encodes portions of the status field of transmitted training frames. k is mapped to the coefficient select echo bits, coef_sts is mapped to the coefficient status bits, and ic_sts is mapped to the initial condition status bit.

SLIP

Causes the next candidate frame marker position to be tested. Repeated invocations of SLIP cause

all possible positions to be evaluated. The precise method for determining the next candidate frame marker position is implementation dependent and beyond the scope of this standard.

UPDATE_C(k)

Updates the value of $c(k)$ based on the current values of the variables k and coef_req . The result of the update is stored in the variable coef_sts . This function implements the algorithm specified in 178B.7.8.4.

UPDATE_IC

Updates all transmit equalizer coefficients based on the current value of the variable ic_req . The result is stored in the variable ic_sts . This function implements the algorithm specified in 178B.7.8.2.

178B.8.3.3 Timers

max_wait_timer

This timer is started when the training control state diagram (Figure 178B–10a) enters the TRAIN_START state. The terminal count of this timer is between N and $N+1$ seconds, where N is the value of the $\text{max_wait_timer_duration}$ variable. The default duration is specified by the clause or annex that defines the interface. When the $\text{max_wait_timer_duration}$ is set to 0, the timer duration is infinite and $\text{max_wait_timer_done}$ always remains false.

propagation_timer

This timer is started when the training control state diagram (Figure 178B–10b) on a lane enters the PATH_READY state. The terminal count of this timer is between 100 ms and 200 ms.

quiet_timer

This timer is started when the training control state diagram (Figure 178B–10a) on a lane enters the QUIET state. The terminal count of this timer is between 100 ms and 200 ms.

recovery_timer

This timer is started when the training control state diagram (Figure 178B–10a) on a lane enters the RECOVERY state. The terminal count of this timer is between 20 ms and 30 ms.

178B.8.3.4 Counters

bad_markers

Count of the number of consecutive frame marker mismatches.

good_markers

Count of the number of consecutive frame marker matches.

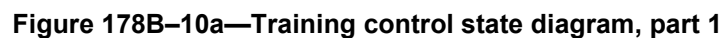
recovery_event_count

This counter increments each time the training control state diagram (Figure 178B–10a) transitions into the RECOVERY state.

178B.8.3.5 State diagram figures

The training control state diagram (Figure 178B–10a and Figure 178B–10b) defines the operation of the ILT function for an interface, and makes use of the per-interface state diagram definitions (178B.8.2) and per-lane state diagram definitions (178B.8.3).

The training control state diagram is supported by the training frame lock, RTS update and coefficient update state diagrams. The training frame lock state diagram (Figure 178B–11) determines when ILT has detected training frame boundaries. The RTS update state diagram (Figure 178B–9) sets the value of the local_rts variable. The coefficient update state diagram (Figure 178B–12) defines the process for updating the local transmitter equalizer coefficients in response to requests from the peer interface.



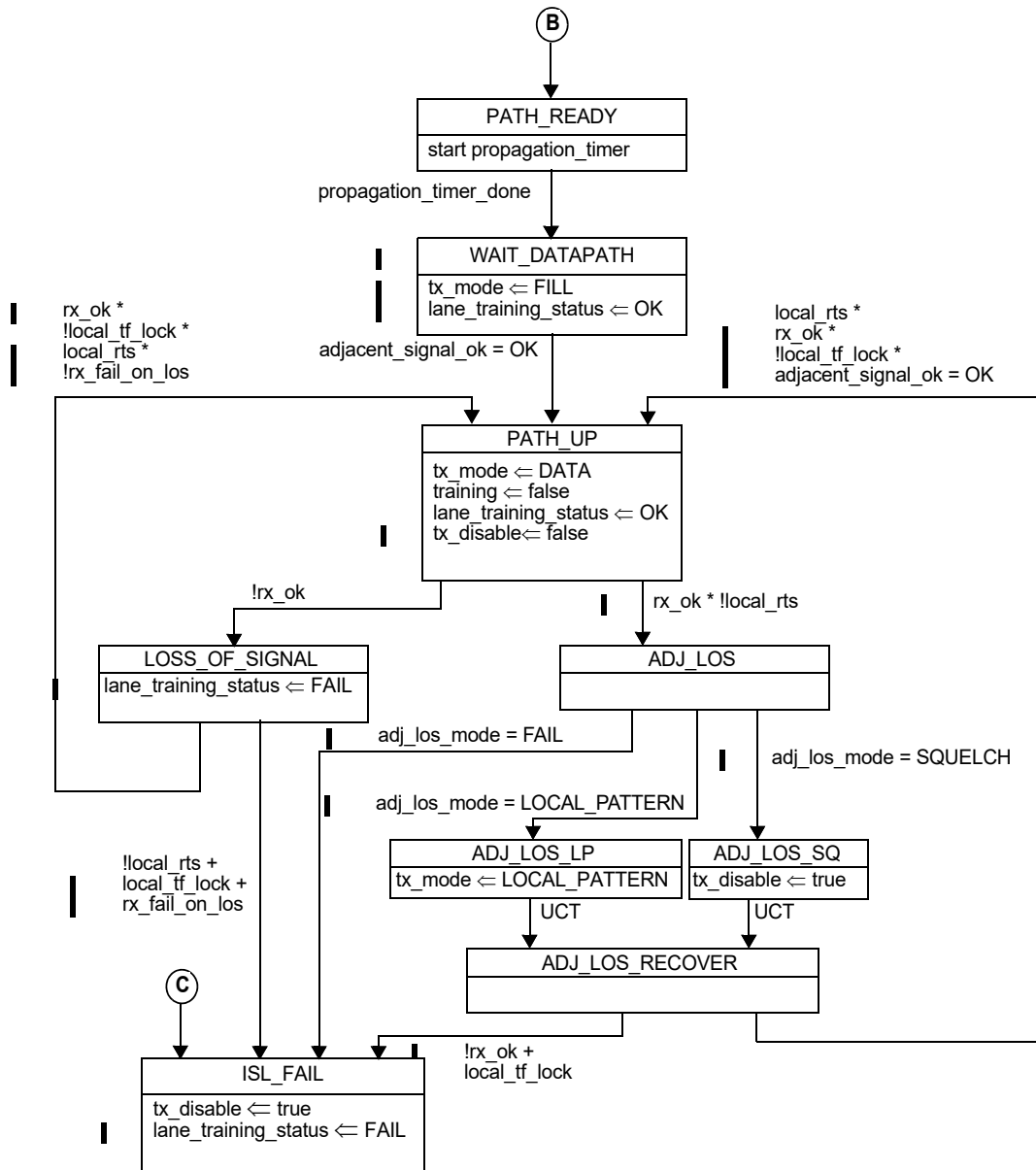


Figure 178B-10b—Training control state diagram, part 2

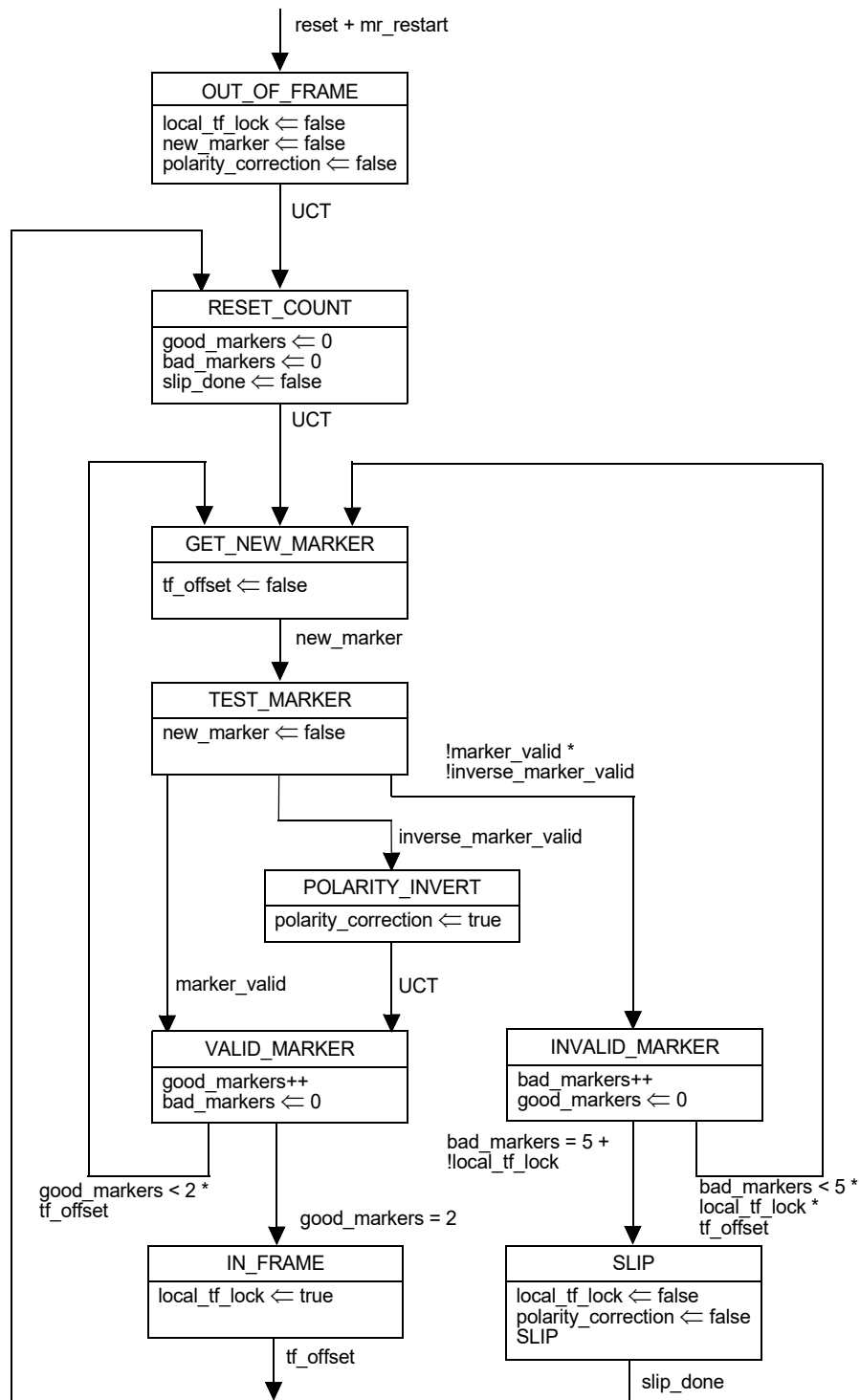


Figure 178B-11—Training frame lock state diagram

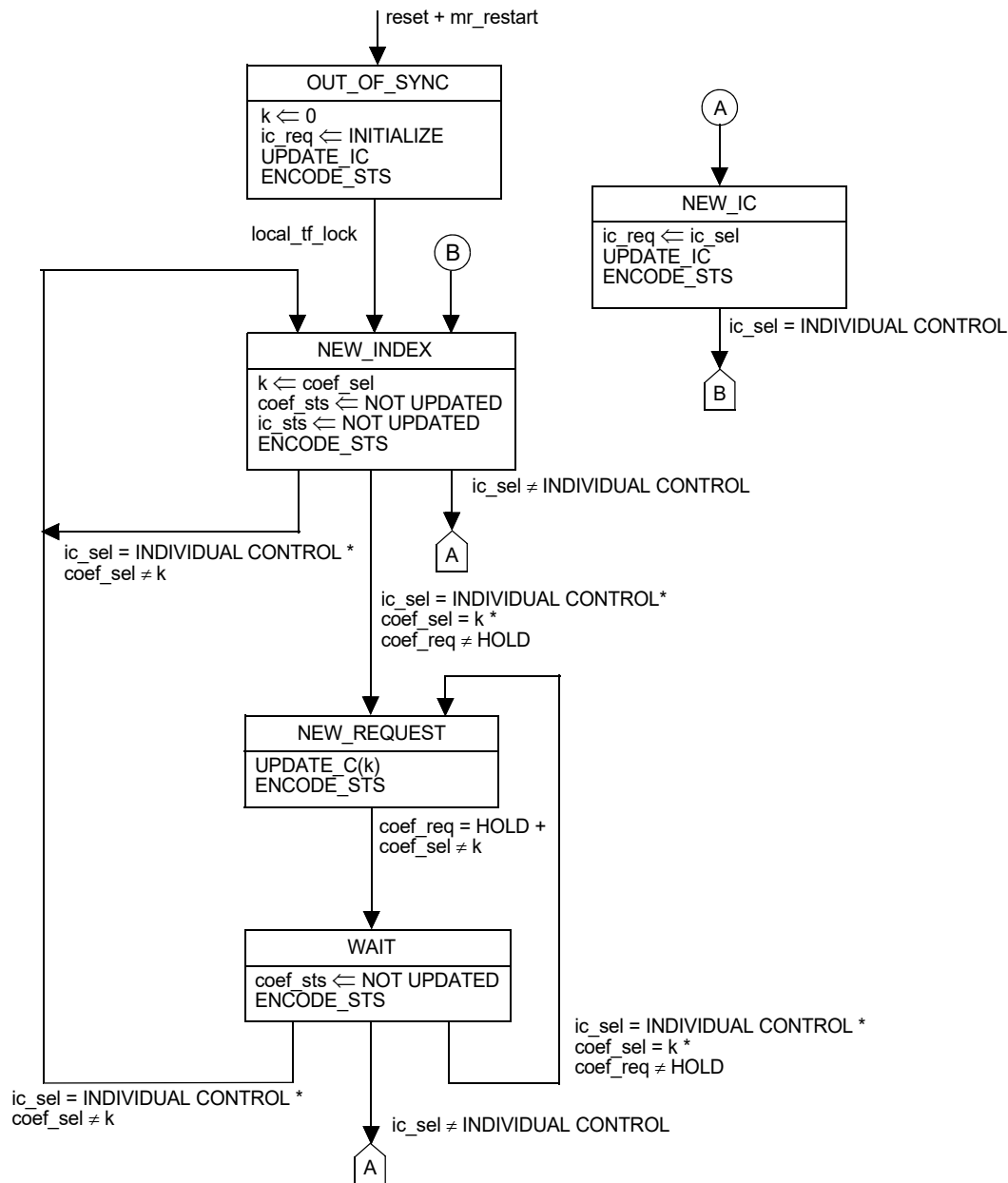


Figure 178B-12—Coefficient update state diagram

178B.9 Training retimers for PAM4 interfaces

The retimer reference model is shown in Figure 178B-13. In normal operation, a retimer passes the data received on one of its interfaces to the other (adjacent) interface using the clock recovered from the received data. During the training process, data might not be available from one interface while training the adjacent interface. The training frames transmission is based on a local clock (USE_TX_CLOCK(local), see 178B.8.2.2) and the transition to the clock recovered by the receiver on the adjacent interface is performed

when the ISL is ready and recovered clock is available as shown in the RTS update state diagram (Figure 178B-9).

The local clock shall meet the signaling rate range specified for the interface.

As shown in the RTS update state diagram (Figure 178B-9), the `local_rts` variable is set to true only after the transmit clock is derived from the clock recovered by the receiver on the adjacent interface such that the transition between clock sources occurs while sending `local_rts` = false.

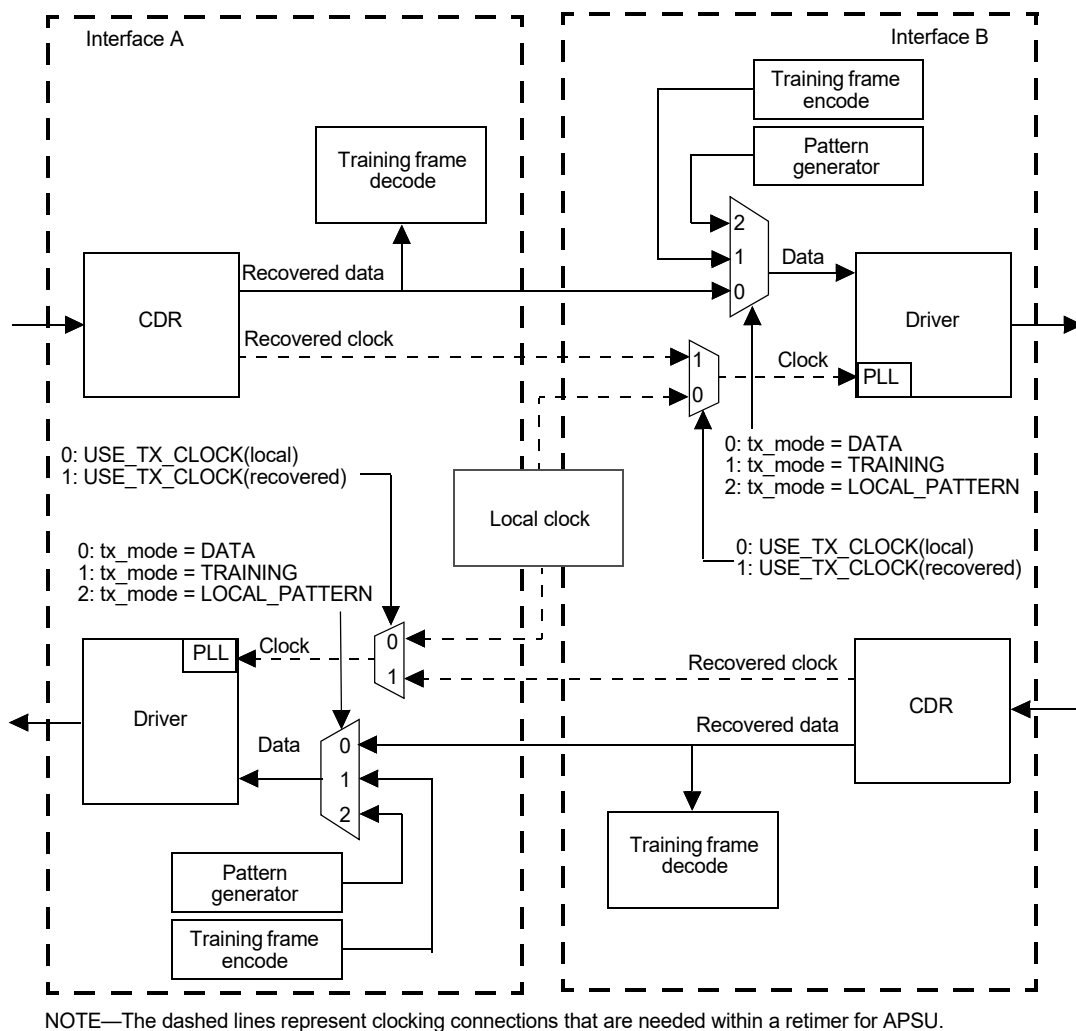


Figure 178B-13—Retimer reference model

178B.10 RTS function for coherent interfaces

The RTS function in coherent interfaces is used to indicate to its peer and adjacent interfaces whether the interface is ready to switch to DATA mode (transmitting data from the client sublayer). The RTS function state diagrams and their associated variables (see 178B.11) define the process used to communicate the interface status and control the clock source and data source transitions.

An overview of the behavior of the RTS function for coherent interfaces is as follows:

- Initially, `c_local_rts` and `c_remote_rts` variables are false.
- Wait for training of the interface to complete (details are PMD-specific) and for the upstream adjacent interface to be ready for DATA mode (the `SIGNAL_OK` parameter from the adjacent interface is set to `READY` or `OK`)
- If the interface is implemented in a retimer that uses the recovered clock in DATA mode, change to the recovered clock.
- When training is complete, the upstream interface is ready, and any necessary clock switch is complete, `c_local_rts` becomes true, indicating readiness to send data (`SIGNAL_OK = READY`).
- Wait for the peer to be ready, as indicated by `c_remote_rts`.
- Wait for `SIGNAL_OK = OK`, and switch to DATA mode (`c_tx_mode = DATA`).

178B.11 State diagrams for coherent interfaces

This subclause is composed of detailed state diagram processes, including the associated definition of variables, functions, counters, and timers. A coherent PMD that squelches the output when the PMD is not ready to send uses the RTS with squelch state diagram (Figure 178B–14). A coherent PMD that does not squelch the output when the PMD is not ready to send uses the RTS without squelch state diagram (Figure 178B–15).

178B.11.1 State diagram conventions

The notation used in the state diagrams follows the conventions of 21.5.

All timers operate as described in 14.2.3.2 with one addition. A timer is reset and stops counting upon entering a state where “stop x_timer” is stated.

Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

178B.11.2 RTS state diagram definitions

A coherent interface implements one instance of either the RTS with squelch state diagram or the RTS without squelch state diagram.

178B.11.2.1 Variables

`c_adjacent_intf_rx_ready`

Boolean variable that is true when `c_adjacent_signal_ok` is either `READY` or `OK` and false otherwise. If there is no adjacent interface this variable is always true.

`c_adjacent_signal_ok`

Enumerated variable derived from the value of the `SIGNAL_OK` parameter on the adjacent service interface. It takes one of the following values: `IN_PROGRESS`, `READY`, `OK`, `FAIL`.

`c_isl_ready`

Boolean variable that is true when `c_local_rx_ready` is true and the interface has completed any other training processes, and is false otherwise. The criteria for completing training are specified by the clause or annex that defines the interface.

`c_local_rts`

Boolean variable that indicates that an interface is ready to send data. It is true if `c_isl_ready` and `c_adjacent_intf_rx_ready` are true and is false otherwise.

c_local_rx_ready	1
Boolean variable that is set to true when the receiver on a lane of the interface has determined that	2
the peer interface transmitter is transmitting the expected signal format, that the local receive	3
equalizers have been optimized, and that no further adjustments are required for normal data	4
transmission. The specific conditions for setting this variable to true or false are implementation	5
dependent. However, it shall be set to false when an input signal corresponding to	6
c_tx_mode = QUIET in the peer is received for 100 ms (the minimum duration of the	7
c_quiet_timer) or longer.	8
	9
c_mr_restart	10
Boolean variable used by management to restart the RTS function.	11
	12
c_remote_rts	12
Boolean variable that indicates the value of c_local_rts variable in the peer interface. The	13
mechanism for signaling RTS status is specified by the clause or annex that defines the interface.	14
For a coherent interface with squelching, c_remote_rts is always true.	15
	16
c_reset	17
Boolean variable that controls the global resetting of the RTS state machine. It is true whenever a	18
reset is necessary including when initiated by reset of the FEC, PMA, or PMD sublayer and during	19
power on, and it is false otherwise.	20
	21
c_rts_status	21
Enumerated variable that indicates the status of the RTS function. It is assigned one of the	22
following values: FAIL, OK, READY, IN_PROGRESS. The values are assigned as defined by the	23
clause or annex that defines the interface.	24
	25
c_tx_mode	26
Enumerated variable that controls the content of the transmitter output of the lane. It is set by the	27
RTS state diagrams (Figure 178B–14 and Figure 178B–15) to one of the values:	28
QUIET: transmitter output squelched	29
TRAINING: transmit the training signal specified by the clause or annex that defines the interface	30
NOT-RTS: transmit the not-ready-to-send signal specified by the clause or annex that defines the	31
interface	32
RTS: transmit the ready-to-send signal specified by the clause or annex that defines the interface	33
FILL: transmit the fill signal specified by the clause or annex that defines the interface	34
DATA: transmit data from the client sublayer	35
	36
c_uses_recovered_clock	37
Boolean variable set to true to indicate that the interface uses a clock recovered by an adjacent	38
interface to drive its output in DATA mode and set to false otherwise.	39

178B.11.2.2 Functions

C_USE_TX_CLOCK(source)	41
Selects the clock source for the transmitter. The source parameter takes one of the values:	42
local: use a local clock to drive the transmitter output	43
recovered: use a clock recovered by the adjacent interface receiver to drive the transmitter output	44

NOTE — The details of the clock recovery and forwarding architecture are beyond the scope of this standard.

178B.11.2.3 Timers

c_forward_clock_timer	49
This timer is started when the RTS state diagram (Figure 178B–14 or Figure 178B–15) enters the	50
SWITCH_CLOCK state. It provides time for the clock switch to propagate to the peer interface.	51
The terminal count of c_forward_clock_timer is between 10 ms and 90 ms.	52

c_propagation_timer

This timer is started when the RTS state diagram (Figure 178B–14 or Figure 178B–15) enters the PATH_READY state. It provides time for the RTS status to propagate before switching to mission data. The terminal count of this timer is between 100 ms and 200 ms.

c_quiet_timer

This timer is started when the RTS state diagram is in the QUIET state. It provides a period of time when the transmitter output is squelched. The terminal count of this timer is between 100 ms and 200 ms.

178B.11.3 State diagram figures

Coherent interfaces that squelch when they are not ready to send shall implement the state diagram shown in Figure 178B–14.

Coherent interfaces that do not squelch when they are not ready to send shall implement the state diagram shown in Figure 178B–15.

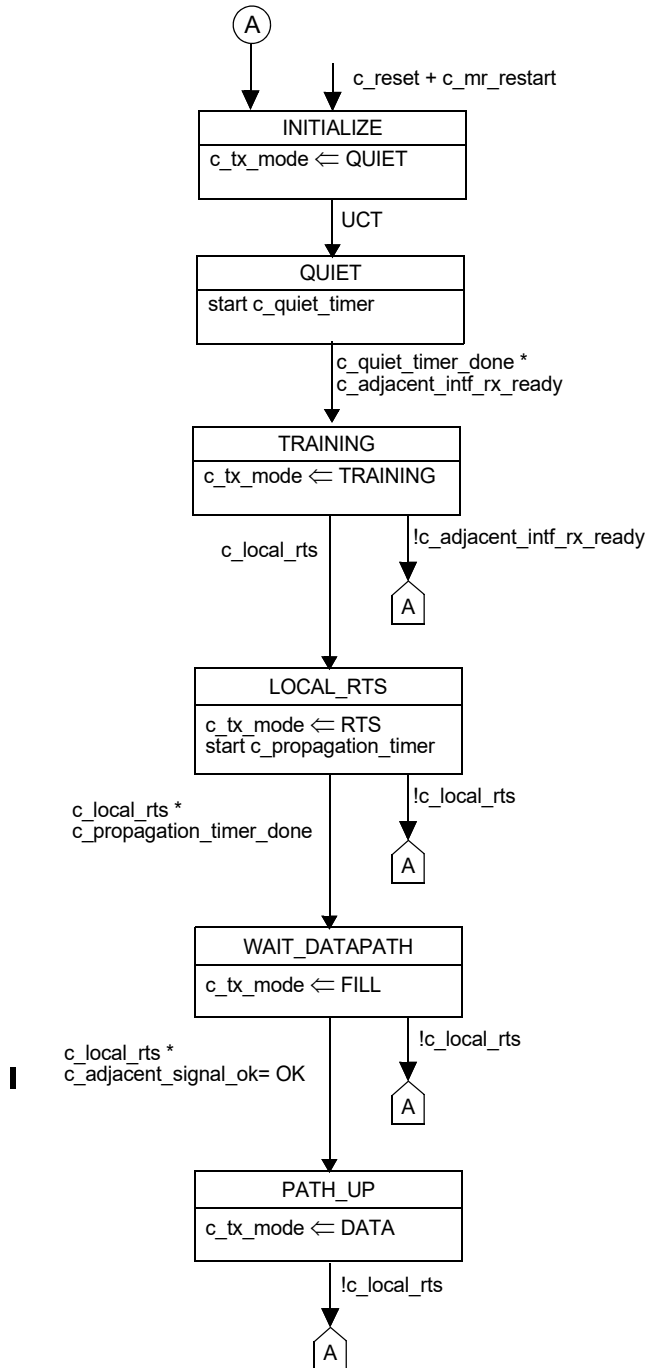


Figure 178B-14—RTS with squelch state diagram

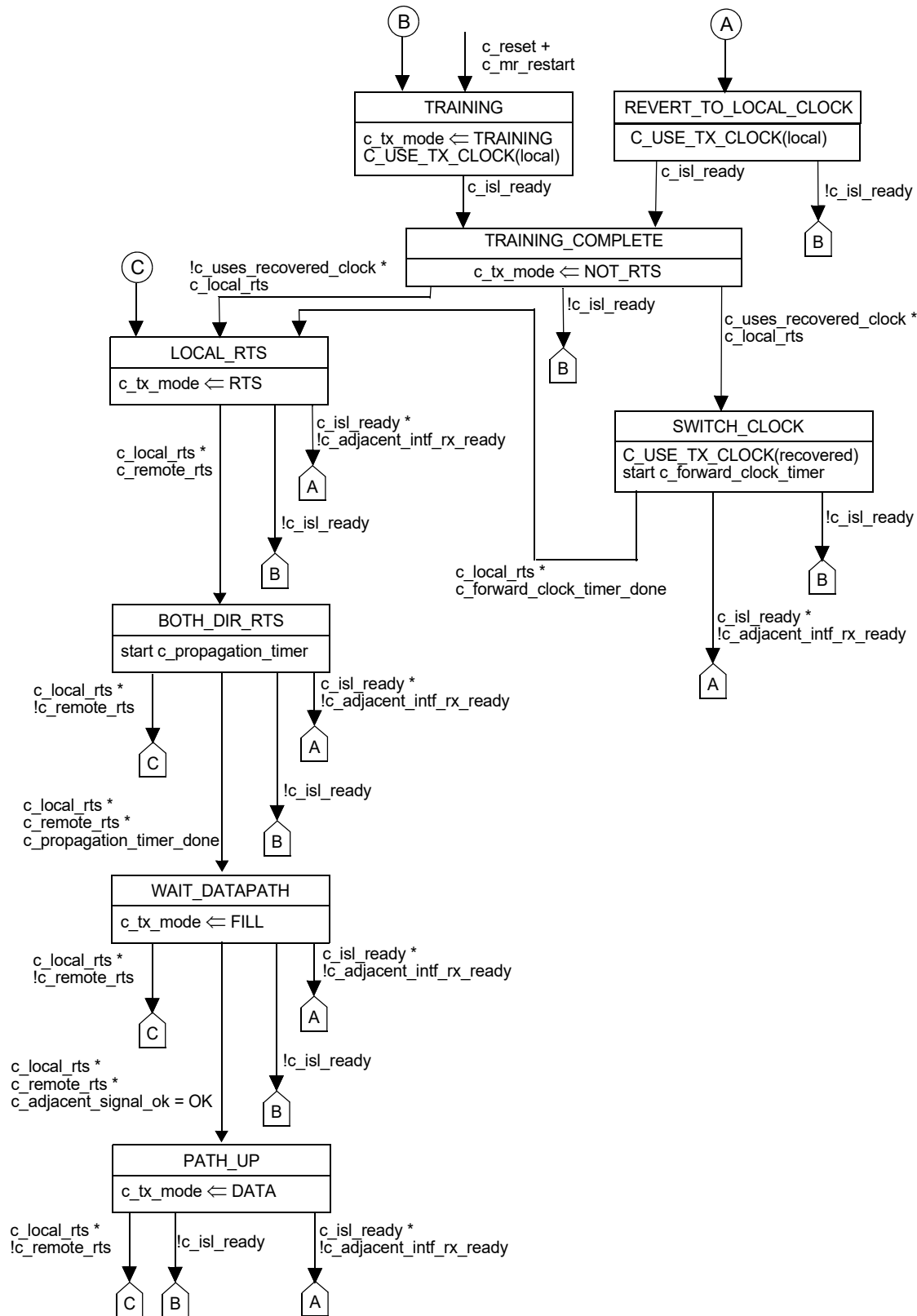


Figure 178B-15—RTS without squelch state diagram

178B.12 Management variables

The control and status variables related with this annex are listed in Table 178B–3 and Table 178B–4.

If the MDIO interface (see Clause 45) is implemented, control and status variables shall be mapped to the MDIO registers as shown in Table 178B–3, Table 178B–4, Table 178B–5 and Table 178B–6.

If the MDIO interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 178B–3—Control variables and MDIO mapping for PAM4 interfaces

Control variable	Variable reference	MDIO register/bit number ^a	MDIO register/bit reference
reset	178B.8.2.1	1.0.15	45.2.1.1
mr_restart	178B.8.2.1	1.150.0	45.2.1.95
mr_training_enable	178B.8.2.1	1.150.1	45.2.1.95
mr_adj_los_mode	178B.8.2.1	1.1479.10:9	45.2.1.169f
identifier_i	178B.7.3.3.1	1.1450.12:11 ^b	45.2.1.168
prbs13_seed_i	178B.7.3.3.1	1.1450.15:14 1.1450.10:0 ^b	45.2.1.168
prbs31_seed_i	178B.7.3.3.3	1.1460.15:0 1.1461.14:0 ^b	45.2.1.169b
ic_req	178B.8.3.1	1.1120.13:11 ^b	45.2.1.161
coef_sel	178B.8.3.1	1.1120.4:2 ^b	45.2.1.161
coef_req	178B.8.3.1	1.1120.1:0 ^b	45.2.1.161
max_recovery_events	178B.8.3.1	1.1479.7:0	45.2.1.169f
max_wait_timer_duration	178B.8.3.1	1.1459.15:0	45.2.1.168a
uses_recovered_clock	178B.8.2.1	1.1479.8	45.2.1.169f
local_mc_mode	178B.7.5.3	1.1420.11:10 ^b	45.2.1.167
local_tp_mode	178B.7.5.2	1.1420.13:12 ^b	45.2.1.167

^a The MDIO register/bit numbers provided in this table apply to a PMD. Mappings of these MDIO register/bit numbers to upper and lower AUI components (see 178B.3) are defined in 176C.8 and 176D.9.

^b Reference is provided for lane 0. Registers for lanes 1 to 7 are located at an offsets from the lane 0 register.

Table 178B-4—Status variables and MDIO mapping for PAM4 interfaces

Status variable	Variable reference	MDIO register/bit number ^a	MDIO register/bit reference
isl_ready	178B.8.2.1	1.1476.8	45.2.1.169c
training_failure	178B.8.3.1	1.151.3 ^b	45.2.1.96
polarity_correction	178B.8.3.1	1.1476.0 ^c	45.2.1.169c
rts_status	178B.8.2.1	1.1476.14:13	45.2.1.169c
training_status	178B.8.2.1	1.1476.11:10	45.2.1.169c
lane_training_status	178B.8.3.1	1.1477.1:0 ^c	45.2.1.169d
training	178B.8.2.1	1.151.2 ^b	45.2.1.96
coef_sts	178B.8.3.1	1.1420.2:0 ^d	45.2.1.167
ic_sts	178B.8.3.1	1.1420.8d	45.2.1.167
local_tf_lock	178B.8.3.1	1.151.1 ^b	45.2.1.96
remote_tf_lock	178B.8.3.1	1.1220.9 ^d	45.2.1.163
rx_ok	178B.8.3.1	1.1478.7:0 ^c	45.2.1.169e
local_rx_ready	178B.8.3.1	1.1420.15 ^d	45.2.1.167
remote_rx_ready	178B.8.3.1	1.1220.15 ^d	45.2.1.163
local_rts	178B.8.2.1	1.1320.10	45.2.1.165
remote_rts	178B.8.2.1	1.1120.10	45.2.1.161
tx_lp_on_adj_los_ability	178B.8.2.1	1.1479.13	45.2.1.169f
tx_sq_on_adj_los_ability	178B.8.2.1	1.1479.14	45.2.1.169f
rx_fail_on_los	178B.8.2.1	1.1479.15	45.2.1.169f

^a The MDIO register/bit numbers provided in this table apply to a PMD. Mappings of these MDIO register/bit numbers to upper and lower AUI components (see 178B.3) are defined in 176C.8 and 176D.9.

^b Bit reference is provided for lane 0. Bits for lanes 1 to 3 are located in the same register and bits for lanes 4 to 7 are located in BASE-R PMD status 2 register (45.2.1.101).

^c Bit reference is provided for lane 0, bits for lanes 1 to 7 are located in the same register.

^d Reference is provided for lane 0. Registers for lanes 1 to 7 are located at offsets from the lane 0 register.

Table 178B–5—Control variables and MDIO mapping for coherent interfaces

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
c_reset	178B.11.2.1	1.0.15	45.2.1.1
c_mr_restart	178B.11.2.1	1.150.0	45.2.1.95
c_uses_recovered_clock	178B.11.2.1	1.1479.8	45.2.1.169f

Table 178B–6—Status variables and MDIO mapping for coherent interfaces

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
c_isl_ready	178B.11.2.1	1.1476.8	45.2.1.169c
c_rts_status	178B.11.2.1	1.1476.14:13	45.2.1.169c
c_local_rx_ready	178B.11.2.1	1.1420.15	45.2.1.167

178B.13 Protocol implementation conformance statement (PICS) proforma for Annex 178B, ILT and RTS functions and support for Autonomous Path Startup for electrical and optical interfaces⁸

178B.13.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 178B, autonomous path startup functions for electrical and optical interfaces, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

178B.13.2 Identification

178B.13.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

178B.13.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Annex 178B, ILT and RTS functions and support for Autonomous Path Startup for electrical and optical interfaces
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
-------------------	--

⁸*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

178B.13.3 Major capabilities

Item	Feature	Subclause	Value/Comment	Status	Support
*MC1	APSU for PAM4 interfaces	178B.5		O/1	Yes [] No []
*MC2	APSU for optical coherent interfaces	178B.10		O/1	Yes [] No []
*MC3	APSU for electrical PAM4 interfaces	178B.5		MC1:O/2	Yes [] No [] N/A []
*MC4	APSU for optical PAM4 interfaces	178B.5		MC1:O/2	Yes [] No [] N/A []
*MC5	APSU for coherent interfaces that squelch	178B.10		MC2:O/3	Yes [] No [] N/A []
*MC6	APSU for coherent interfaces that do not squelch	178B.10		MC2:O/3	Yes [] No [] N/A []
*MD	MDIO	178B.12		O	Yes [] No []

178B.13.4 ILT and RTS functions for PAM4 ISLs

Item	Feature	Subclause	Value/Comment	Status	Support
IL1	Frame marker	178B.7.3.1	Transmit frame marker	MC1:M	Yes [] N/A []
IL2	Polarity detection	178B.7.7	Detect and correct polarity	MC1:M	Yes [] N/A []
IL3	Synchronous PRBS13 training pattern	178B.7.3.3.1	Each lane implements four generator polynomials defined in Table 178B–1	MC1:M	Yes [] N/A []
IL4	Training pattern seed	178B.7.3.3.1	State set to the value of prbs13_seed_i at the start of the training pattern	MC1:M	Yes [] N/A []
IL5	PRBS13 free running training pattern	178B.7.3.3.2	Each lane implements four generator polynomials defined in Table 178B–1	MC1:M	Yes [] N/A []
IL6	PRBS31 free running training pattern	178B.7.3.3.3	Each lane implements the PRBS31 generator	MC1:M	Yes [] N/A []
*IL7	E1 format	178B.7.4	E1 format used	MC3:M	Yes [] N/A []
*IL8	O1 format	178B.7.4	O1 format used	MC4:M	Yes [] N/A []
IL9	E1 format control field structure	178B.7.4	As shown in Table 178B–2	IL7:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
IL10	O1 format control field structure	178B.7.4	As shown in Table 178B–3	IL8:M	Yes [] N/A []
IL11	E1 format status field structure	178B.7.5	As shown in Table 178B–4	IL7:M	Yes [] N/A []
IL12	O1 format status field structure	178B.7.5	As shown in Table 178B–5	IL8:M	Yes [] N/A []
IL13	Training frame lock	178B.7.6	As shown in Figure 178B–11	MC1:M	Yes [] N/A []
IL14	Initial condition setting request process	178B.7.8.1	When requested, follow initial condition setting request process	IL7:M	Yes [] N/A []
IL15	Initial condition setting response process	178B.7.8.2	Follow initial condition setting response process	IL7:M	Yes [] N/A []
IL16	Coefficient update request process	178B.7.8.3	When requested, follow coefficient update request process	IL7:M	Yes [] N/A []
IL17	Coefficient update response process	178B.7.8.4	Follow coefficient update response process	IL7:M	Yes [] N/A []
IL18	Handshake timing	178B.7.10	When the transmitted frame lock bit is 1, acknowledge requests within less than 2 ms	MC1:M	Yes [] N/A []
IL19	Local clock rate	178B.9	Retimer local clock signaling rate range as specified for the interface.	MC1:M	Yes [] N/A []

178B.13.5 State diagrams

Item	Feature	Subclause	Value/Comment	Status	Support
SM1	RTS update state diagram	178B.8.2	As shown in Figure 178B–9	MC1:M	Yes [] N/A []
SM2	Per-lane state diagrams	178B.8.3	As shown in Figure 178B–10a, Figure 178B–10b, and Figure 178B–12	MC1:M	Yes [] N/A []
SM3	RTS with squelch state diagram	178B.11	As shown in Figure 178B–14	MC5:M	Yes [] N/A []
SM4	RTS without squelch state diagram	178B.11	As shown in Figure 178B–15	MC6:M	Yes [] N/A []

178B.13.6 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Alternate access to management variables is provided	178B.12		!MD:M	Yes [] N/A []
M2	Mapping of MDIO control and status variables for PAM4 interfaces	178B.12	Table 178B–3 and Table 178B–4	MD*MC1:M	Yes [] N/A []
M3	Mapping of MDIO control and status variables for coherent interfaces	178B.12	Table 178B–5 and Table 178B–6	MD*MC2:M	Yes [] N/A []